Dataset Details & Exploratory Analysis Report

AI-Powered Restaurant Operations Assistant — Dataset Engineering Specification

1. Dataset Overview (Name, Source, Link & Format)

Dataset Name:	Yelp Open Academic Dataset
Dataset Source:	Yelp Dataset Challenge (Official Academic Research Program)
Dataset Link:	https://www.yelp.com/dataset
Data Format (Raw):	JSON Lines (.json files per entity: business.json, review.json, user.json, tip.json)
Data Format (Storage):	SQLite (restaurant_operations.db), ChromaDB HNSW Vector Store, and CSV export
Primary Join Key:	**business_id** (UUID linking reviews to physical restaurant profiles)

2. Key Fields Specification (Raw & Enriched Operational Schema)

Entity	Field Name	Data Type	Description & Operational Purpose
Business	business_id	string (UUID)	Primary key uniquely identifying the physical restaurant establishment.
Business	name	string	Official restaurant trade name.
Business	categories	string (CSV)	Industry & cuisine tags (used for domain filtering).
Business	stars	float (1-5)	Historical weighted aggregate star rating.
Review	review_id	string (UUID)	Unique review identifier.
Review	text	string (Text)	Unstructured dining narrative containing complaints and feedback.
Review	stars	integer (1-5)	Star rating assigned by customer (1-2 = negative complaint).
Review	date	datetime	ISO timestamp used for Prophet time-series trend projections.
Enriched	clean_text	string	Normalized text stripped of HTML entities and non-printable noise.
Enriched	operational_category	string	Classified into 8 operational areas (Food Quality, Service, Wait Time, etc.).
Enriched	severity_score	float (1-5)	Computed operational friction severity based on negative cues.
Enriched	priority_score / tier	float / string	Composite priority (1-100) mapped to High/Medium/Low SLA routing.
Enriched	embedding_vector	float[384]	Dense sentence representation via all-MiniLM-L6-v2 in ChromaDB.

3. Dataset Purpose & Operational Value

The Yelp Open Academic Dataset serves as the real-time operational sensor across 1,184 restaurants. It overcomes **Vanity Metric Blindness** (where a 4.2-star average obscures Friday night kitchen bottlenecks) by enabling:

- **Hybrid RAG Semantic Search:** Finding historical operational incidents using manager queries (e.g., 'patio drink delay').
- **Root Cause Diagnostics & Multi-Agent Orchestration:** Feeding the 5-Agent LangGraph workflow to produce actionable SOP playbooks.
- **Predictive Time-Series Projections:** Training Facebook Prophet models to forecast seasonal complaint surges.

4. Data Preprocessing Performed

1. **Domain Taxonomy Filtering:** Applied strict regex rules over business categories to isolate authentic restaurant venues.
2. **Entity Resolution & Joining:** Merged reviews with business profiles using foreign key business_id.
3. **Text Normalization & Noise Cleaning:** Unescaped HTML characters, stripped control codes, collapsed irregular whitespaces.
4. **Semantic Chunking:** Whole-review boundary preserving chunking (mean length ~135 tokens) ensuring complete narrative integrity.
5. **Operational Categorization & Feature Scoring:** Automatic tagging into 8 operational categories; calculated Severity (1-5) and Priority (1-100).
6. **Vector Embedding Generation:** Generated dense 384-dimensional embeddings via all-MiniLM-L6-v2 indexed in ChromaDB.

5. Exploratory Data Analysis (EDA) Summary

Metric	Observed Value in Ingestion Sample
Total Businesses Scanned	5000
Filtered Restaurant Establishments	2272

Reviews Linked to Restaurants	2723
Average Restaurant Star Rating	3.54 / 5.0
Top Categories Distribution	Restaurants (1762), Food (917), Nightlife (386), Bars (373), Sandwiches (279)
Review Rating Distribution	1 Stars: 263, 2 Stars: 204, 3 Stars: 321, 4 Stars: 756, 5 Stars: 1179